# Qizhou Deng

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Portfolio: qizhoud.github.io/qizhoudeng.github.io · GitHub: github.com/QizhouD

## RESEARCH INTERESTS

Game AI and Computational Media: motion and trajectory generation for interactive digital humans; multimodal player modeling and performance analytics in games; search-based Game AI and procedural content generation.

## EDUCATION BACKGROUND

**Queen Mary University of London** 09/2022 – 09/2023
M.Sc. in Computer Games

**University of Liverpool** 09/2020 – 07/2022
B.S. in Computer Science

**Xi'an Jiaotong-Liverpool University** 09/2018 – 07/2020
B.S. in Information and Computing Science

## RESEARCH AND SELECTED PROJECTS

**Language-conditioned generative trajectory planning for real-time digital humans** 2026 – Present
Repo: github.com/QizhouD/language-conditioned-generative-trajectory-planning-for-real-time-digital-humans

- Research MVP for short-horizon, language-conditioned 2D root-trajectory planning for interactive digital humans: five intents (idle, walk, turn, stop, wave), candidates sampled at **10 Hz**, collision-aware rejection and ranking, and rolling replanning.
- Implemented a **conditional flow-matching** model and a **deterministic Transformer baseline** on synthetic navigation scenes; the baseline detours around obstacles with a collision loss and produces straighter empty-scene paths with path supervision (figures on the portfolio site and in the repo).
- Built the Unity embodiment layer (github.com/QizhouD/LLM-NPC): a language model tags each reply with one of the five intents, which becomes a plan request that drives the character's root at 10 Hz through the planner's trajectory contract; plans currently come from a procedural mock source, not the trained model.
- Next: stabilizing flow-matching trajectories, a learned plan source for the Unity embodiment, and full-body motion.

**AI-FPS-Coach: offline CS2 practice review** 2026 – Present
Repo: github.com/QizhouD/AI-FPS-Coach

- Designed an offline practice-review pipeline: OBS capture, CUDA YOLO analysis after the session, and a Unity war room that replays the footage with detection overlays and a session-level, **FOV-aware** aim report (crosshair deviation and tracking; deviation at fire, reaction time and overcorrection from audio).
- Validated target selection against **44** server-confirmed headshots on range footage: median error at the shot of **0.36°** when the head is placed from the body box, vs **2.48°** when ranking detected head boxes only.

**Improved MCTS agent in Pommerman** 03/2023 – 06/2023
**Queen Mary University of London | Supervisor: Dr Diego Perez-Liebana**
Repo: github.com/QizhouD/java-pommerman/tree/master/ImprovedMCTS

- Compared basic AMAF, **α-AMAF** and RAVE enhancements to MCTS in the Java Pommerman framework; α-AMAF raised the win rate by about **5 percentage points** over the benchmark MCTS agent under full observability and was on par under partial observability.

**Modeling player navigation in games (critical review)** 03/2023 – 06/2023
**Queen Mary University of London | Supervisors: Dr Jeremy Gow, Dr Laurissa Tokarchuk**

- Surveyed GUR navigation analytics and pathfinding (A*, DPA hybrids, RRT, etc.); case study on The Last of Us NPC AI (NavMesh, exposure maps, sensory models); identified gaps in sim-to-real validation and audiovisual cues.

**Smoking-behaviour detection with YOLOv5 (MSc project)** 03/2023 – 09/2023

**Queen Mary University of London | Supervisors: Dr Jeremy Gow, Dr Haowei Peng**

- Collected and annotated (LabelImg) a **783-image** smoking-behaviour dataset; trained YOLOv5 in PyTorch and compared it with YOLOv4 on a 9:1 split (**mAP 93.6% vs 91.3%**; 50 vs 37 FPS on an RTX 3070 laptop GPU); deployed a PyQt5 desktop demo.
- This project strengthened the computer vision foundation later used in AI-FPS-Coach

**Dwarf & Fire: C++17 / SFML action game** 12/2022 – 03/2023; extended 2025 – 2026

**Queen Mary University of London coursework | Instructor: Dr Diego Perez-Liebana**

Repo: github.com/QizhouD/2D-action-game

- Component-based entities run through a single fixed-order system pipeline (input, AI, movement, TTL) on a fixed 60 Hz timestep; Command, Observer and Service Locator patterns; unit tests and CI on Windows and Linux.

## WORK EXPERIENCE

**Software Engineer, China Telecom Ltd., China** 03/2024 – Present

- Deployed **on-prem LLM + RAG** for context-aware internal data analysis and querying.
- Built a **target recognition** microservices pipeline (deep-learning detection, Flask, Docker, RabbitMQ) for high-throughput video streams.
- Developed Spring Boot / Vue payroll and approval workflows for production systems.

**Operations Assistant Intern, China Telecom (Europe) Ltd., London, UK** 03/2023 – 05/2023

- Python (Pandas / Matplotlib) analytics and VBA automation for e-commerce reporting.

## SKILLS

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| **Programming Languages** | C# (Proficient) \| Java (Proficient) \| Python (Proficient) \| C/C++ (Familiar) |
| **Backend Development** | Spring Boot \| JUnit \| RESTful APIs \| MySQL \| Git \| .NET |
| **AI/ML Skills** | TensorFlow \| PyTorch \| Pandas \| NumPy \| OpenCV \| Matplotlib |
| **Frontend Exposure** | JavaScript \| Vue.js \| HTML/CSS |
| **Tools & Platforms** | Docker \| Linux \| Postman \| Unity |